

Daniel Ray Edgar

Chief Technology Officer | AI Systems Engineer | Researcher

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PROFESSIONAL SUMMARY

AI systems engineer and founder building applied artificial intelligence for the highest-stakes work in finance. Currently **Chief Technology Officer of Finsider**, automating Quality of Earnings — the mandatory financial-diligence report behind every M&A transaction. Studied **Computer Science at Queen's University** before leaving to build; **raised \$230,000 at a \$2.3M valuation** as co-founder and CEO of an Antler Canada-backed company at 19. Author of original, single-author research on the reliability of large-language-model reasoning, validated against four frontier models. Designs the theory, builds the systems, and engineers the autonomous infrastructure that runs them — with auditability and supervised autonomy as first principles.

This document is both a technical ability overview and the project/role itinerary for the applicant. Every system referenced links to its public source repository.

CORE TECHNICAL COMPETENCIES

Languages	Python, TypeScript / JavaScript, Swift (iOS), SQL, Bash, L ^A T _E X
Applied AI / LLMs	LLM orchestration; Retrieval-Augmented Generation (Vertex AI RAG, corpus-per-workload, multi-query retrieval, dedup/ranking); structured extraction with calibrated confidence scoring; agentic workflows and multi-agent systems; process reward models; frontier-model evaluation (GPT-5, Claude Opus/Sonnet 4.6, Gemini 3.1 Pro); Anthropic, Google Vertex AI, and Gemini 2.0 Flash APIs
Agentic systems	Supervised-autonomy architectures; evidence-gated delivery pipelines; ticket-native state machines; acceptance-contract prompting; Playwright-based evidence capture; process supervision and watchdogs; safe automated Git operations
Full-stack	Next.js 16, React 19, TypeScript, Tailwind CSS v4, shadcn/ui; Supabase (PostgreSQL, Row-Level Security, Auth, Storage), Firebase, Sanity (headless CMS); Stripe, Resend, Twilio; SSE streaming, async job queues, webhook pipelines
Cloud & infrastructure	Google Cloud Platform (Compute Engine, Cloud Storage, Vertex AI), Vercel; self-hosted Proxmox cluster, Docker, systemd , macOS launchd , Tailscale; CI-aware multi-repo delivery
Applied math / research	Probability theory, Bernoulli/Hoeffding bounds, maximum-likelihood estimation, the delta method, confidence-interval construction; empirical model evaluation
Financial domains	M&A Quality of Earnings; Canadian mortgage underwriting (GDS/TDS, OSFI guidelines, stress testing); financial-document AI; document-authenticity and fraud-signal detection

ORIGINAL RESEARCH CONTRIBUTION

Uncertainty Propagation in Tree-Structured Language Model Reasoning (github.com/dm3n/uncertainty-propagation)

Single-author paper formalizing how per-step errors compound across multi-step LLM reasoning. Introduces the **Reasoning Tree Model (RTM)**, which unifies chain-of-thought, self-consistency, and tree-of-thought as special cases, and derives the governing results from first principles:

- **Chain success** = $\prod_i (1 - \theta_i)$, decaying exponentially in depth (Theorem 3.1).
- **Tree majority vote** decays only *sub*-exponentially in compute, $\exp(-\Omega(N^{1-1/(d+1)}))$ — a qualitatively different reliability regime (Theorem 4.2).
- Closed-form **compute-optimal branching factor and depth** (Theorem 4.3); MLE for step-failure rates with delta-method confidence intervals on end-to-end success (Theorems 5.1–5.2).
- **Empirical validation** against GPT-5, Claude Opus 4.6, Claude Sonnet 4.6, and Gemini 3.1 Pro on GSM8K, MATH, and MMLU-Pro: predictions track observed accuracy to within **~1% mean absolute error**. Signature result: at a fixed 48-step compute budget, a chain succeeds $\approx 0.2\%$ of the time while a depth-3 tree succeeds **97.9%**.

This research is the design theory behind the production systems below: decompose work into short, independently verified, tree-structured units rather than one long, exponentially decaying chain.

FLAGSHIP SYSTEMS

Symphony — autonomous software-delivery runner (github.com/dm3n/symphony)

A Jira-native, evidence-gated, supervised-autonomy runner that drives a product ticket from backlog to a reviewed, evidence-backed pull request. **Built as CTO of Finsider and deployed live in its engineering environment.** ~2,300-line standard-library Python orchestrator.

- **Jira as the state machine** — ticket status and labels are the control flow; no external queue or database; crash-safe because state is re-derived from Jira each poll.
- **Per-ticket isolated Git workspaces**; acceptance-contract prompting of a coding agent; **Playwright** screenshot/video evidence capture.
- **Intent-classified human review** (question vs. rework vs. approval); **cross-repo scope critic**; **evidence-integrity gating**; **-force-with-lease** safe Git; recursive process-tree cleanup; bounded failure auto-retry.
- Dual deployment: macOS `launchd` (local) and an always-on GCP Compute Engine VM under `systemd`.

Macintosh — multi-agent engineering operating system (github.com/dm3n/macintosh)

A reproducible, self-hosted “engineering OS”: a six-stage agentic coding pipeline (Plan → Code → Test → QA → Review) implemented as Claude Code skills atop the Superpowers discipline; a unified four-agent setup (Claude Code, Codex, Gemini, OpenCode) sharing context; a Personal Knowledge Brain (Obsidian + LLM compiler); and a Proxmox/Docker homelab with a Kali AI node and a human-gated approval model.

AI PLATFORMS — AIRBANK (github.com/dm3n/airbank, github.com/dm3n/airbank-mortgage-platform)

Quality-of-Earnings engine

An 11-section, RAG-grounded extraction pipeline producing analyst-grade QoE workbooks where every number is traceable to a source document.

- Corpus-per-workbook Vertex AI RAG with parallel multi-query retrieval and content-fingerprint dedup; hybrid *discovery mode* (fuzzy anchor-row matching + free row discovery) robust to non-standard financial statements.
- Audit-quality citation (document, page, verbatim excerpt per cell); calibrated confidence (0.4–1.0) engineered to *flag missing data rather than fabricate*; deterministic fast path that bypasses RAG

for clean spreadsheets.

- Cross-section reconciliation via a formula resolver; live SSE analysis streaming.

Mortgage underwriting platform

Full-lifecycle AI origination for Canadian brokers: Canadian-specific document extraction (T4, NOA, pay-stub, LOE), **fraud-signal detection** (PDF metadata, font inconsistency, name/income mismatch), GDS/TDS engine with stress-test rate, one-call intake orchestration, real-time upload verification, and autonomous cron-driven pipeline orchestration generating role-aware priorities.

SELECTED PRODUCTS

- **VisionClaw** (github.com/dm3n/VisionClaw) — real-time AI vision assistant for Meta Ray-Ban glasses (Swift/SwiftUI, Meta Wearables DAT SDK, Gemini Live over WebSocket). Accessibility-first: hands-free Siri activation, resilient streaming, proactive spatial narration for visually impaired users.
- **ROGI** (github.com/dm3n/rogi, github.com/dm3n/rogi-v2) — AI-native mortgage intake funnel with a Canadian qualification engine (GDS/TDS, stress test, CMHC brackets). First version sold for \$7,500.
- **Nassau** (github.com/dm3n/nassau) — complete golf e-commerce platform built from scratch: Next.js + Sanity headless CMS + Stripe payments + Firebase + React Query, with premium product UX.
- **Sacred Secretion Agent** (github.com/dm3n/sacred-secretion-agent) — fully autonomous lunar-cycle email agent (local astronomical computation, trigger-based sequencing, Vercel cron).
- Additional: **ventureroadmap**, github.com/dm3n/nodebase, github.com/dm3n/ratestore, github.com/dm3n/human-divinity (second long-form research manuscript).

ITINERARY — ROLES, PROJECTS & TECHNICAL OBJECTIVES

Primary location of work: Toronto, Ontario, Canada (Antler Canada / OneEleven, 325 Front St W).

Co-founder & CEO — Flagpost AI Inc. (Sep 2025 – Q1 2026)

- Selected into the **Antler Canada TOR8 residency** (commenced Sep 2025) from thousands of applicants; passed the Antler Investment Committee.
- **Funding milestone:** raised \$230,000 at a \$2.3M valuation via a SAFE from Antler Canada Fund I, L.P.; incorporated Flagpost AI Inc. (Nov 2025).
- **Duties:** company formation, team and cap-table structure, product direction (AI-native sell-side M&A preparation), investor relations.
- **Outcome:** stepped aside to pursue the broader M&A-intelligence thesis directly.

Founder — Airbank (Q1 2026)

- **Technical objectives:** design and ship an AI-native operating system for M&A, beginning with the Quality-of-Earnings wedge; build the QoE and mortgage-underwriting platforms above.
- **Product development phases:** RAG ingestion → section-wise extraction with confidence flags → auditable source-linked cells → reconciliation/export; mortgage intake → AI verification → underwriting audit → pipeline orchestration.
- **Customer acquisition:** secured signed LOIs, NDAs, and advisory agreements with senior professionals across private credit, M&A advisory, investment banking, and private equity (design-partner network).

Chief Technology Officer — Finsider (2026 – Present)

- **Compensation:** \$200,000 annual salary (CTO) — remuneration materially above the norm for the role and tenure.
- **Technical objectives:** automate Quality of Earnings end-to-end; own the multi-repo platform (frontend, backend engine, AI/agent services) and the financial-correctness layer (Balance Sheet, P&L, Cash Flow, Cash Proof tie-out and standardized export).
- **Signature deliverable:** designed and deployed **Symphony** (above) as the company’s agentic delivery infrastructure.
- **Forward plan (petition period):** (1) harden and scale the QoE automation engine and agentic delivery pipeline; (2) expand financial-statement correctness and export coverage; (3) grow enterprise/customer adoption of the automated QoE product; (4) advance the supervised-autonomy research-to-production loop.

EDUCATION

Queen’s University — Computer Science. Left to build full-time after raising venture funding.

REPOSITORY INDEX

All systems referenced are public unless marked private.

Portfolio (this overview)	github.com/dm3n/portfolio
Symphony	github.com/dm3n/symphony
Macintosh (engineering OS)	github.com/dm3n/macintosh
Research — reasoning	github.com/dm3n/uncertainty-propagation
Research — manuscript	github.com/dm3n/human-divinity
Airbank QoE	github.com/dm3n/airbank <i>(private)</i>
Airbank Mortgage	github.com/dm3n/airbank-mortgage-platform <i>(private)</i>
ROGI	github.com/dm3n/rogi github.com/dm3n/rogi-v2
VisionClaw	github.com/dm3n/VisionClaw
Nassau	github.com/dm3n/nassau
Sacred Secretion Agent	github.com/dm3n/sacred-secretion-agent
Other	github.com/dm3n/nodebase github.com/dm3n/ratestore github.com/dm3n/OpenCV

Prepared June 26, 2026. Sensitive investor, customer, and financial specifics are maintained in a private data room available on request.